

多雲端資料串流

來源：<https://codelabs.developers.google.com/codelabs/Multi-Cloud%20Data%20Streaming?hl=zh-tw>

步驟數：3

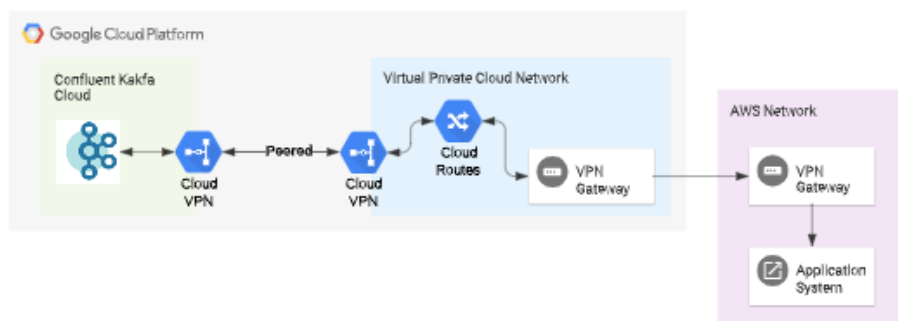
在本程式碼研究室中，您將在 GCP 上設定跨區域的 Confluent Kafka，其他超大規模雲端服務 (例如 AWS) 也能透過 VPN 通道存取，提供真正的多雲串流功能，來源位於 AWS，目標位於 GCP。

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1. 簡介



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為何需要多雲端串流？

企業正積極尋求在多個雲端平台分散工作負載，因為在企業的營運工作流程中採用多個雲端服務供應商，可帶來多項優勢，例如：

- 受制於相同廠商
- 降低營運持續風險
- 依附單一技術堆疊

建構項目

在本程式碼研究室中，您將利用 Confluent Cloud，在 AWS 和 GCP 之間設定多雲端資料串流。您將學會以下內容：

- 在 GCP 上的 Confluent Cloud 中設定專屬 Kafka 叢集。
- 在 Confluent Cloud 和 GCP 專案之間設定虛擬私有雲對等互連，以便在私人網路上存取。
- 在 AWS 專案和 GCP 專案之間設定 VPN 通道，讓兩個超大規模運算服務的資源可以連線。
- 從 AWS 存取 Confluent Kafka，並利用 Confluent Kafka Cloud 設定從 AWS 到 GCP 的串流管道。

課程內容

- 如何設定 Confluent Cloud Kafka 與 GCP 專案之間的 VPC 對等互連。
- 如何在 AWS 和 GCP 專案之間設定 VPN 通道。
- 如何從 AWS 專案存取 GCP 代管的 Confluent Cloud。

本程式碼研究室著重於建構多雲端資料串流。我們不會對與主題無關的概念和程式碼多做介紹，但會事先準備好這些程式碼區塊，屆時您只要複製及貼上即可。

軟硬體需求

- AWS 專案存取權。
 - GCP 專案存取權。
 - GCP 和 AWS Cloud 的經驗。
 - 透過 GCP Marketplace 訂閱 Confluent Kafka。
-

2. 開始設定

設定 AWS 與 GCP 之間的 VPN 連線

- 本示範使用 GCP 中的預設虛擬私有雲 (asia-southeast1)，以及 AWS 中的 ng-vpc-103-mum (ap-south1)。

VPC > Subnets > subnet-0c8006b742232a47b

subnet-0c8006b742232a47b / ng-sub-mum-103-2b

Actions

Details

Subnet ID subnet-0c8006b742232a47b	Subnet ARN arn:aws:ec2:ap-south-1:233797219673:subnet/subnet-0c8006b742232a47b	State Available	IPv4 CIDR 10.120.2.0/24
Available IPv4 addresses 250	IPv6 CIDR -	Availability Zone ap-south-1a	Availability Zone ID aps1-az1
VPC vpc-03469cf4ed658e69d ng-vpc103-mum	Route table rtb-0544c7eeaf4bc7035 ng-rt-103	Network ACL acl-0d75b60dcfd08fc4	Default subnet No
Auto-assign public IPv4 address No	Auto-assign IPv6 address No	Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
Outpost ID -	IPv4 CIDR reservations -	IPv6 CIDR reservations -	IPv6-only No
Hostname type IP name	Resource name DNS A record Disabled	Resource name DNS AAAA record Disabled	Owner 233797219673

←

Subnet details

✎

EDIT

🗑

DELETE

default

VPC Network

default

Region

asia-southeast1

IP address range

10.148.0.0/20

Gateway

10.148.0.1

Private Google Access

Off

Flow logs

Off

[View flow logs](#)

EQUIVALENT REST

- 在 AWS 上建立客戶閘道和虛擬私有閘道 (VPG)。將 VPG 附加至 AWS VPC。

Create Customer Gateway

Actions

search : 103

Add filter

	Name	ID	State	Type	IP Address	BGP ASN
	ng-aws-103-gw	cgw-00cc6d655e6407a0b	available	ipsec.1	34.126.140.162	65000

Create Virtual Private Gateway

Actions

search : 103

Add filter

1 to 1 of 1

	Name	ID	State	Type	VPC	ASN (Amazon side)
	ng-aws-vpg-103	vgw-08db0cb9576b8417b	attached	ipsec.1	vpc-03469cf4ed658e69d ng-vpc103-mum	64512

Virtual Private Gateway: vgw-08db0cb9576b8417b

Details

Tags

ID

vgw-08db0cb9576b8417b

State

attached

Type

ipsec.1

VPC

vpc-03469cf4ed658e69d | ng-vpc103-mum

ASN (Amazon side)

64512

- 在 AWS 上建立站對站 VPN 連線。

Create VPN Connection

Select the target gateway and customer gateway that you would like to connect via a VPN connection. You must have entered the target gateway information already.

Name tag

ng-aws-103-gw

Target Gateway Type

☒ Virtual Private Gateway

☐ Transit Gateway

☐ Not Associated

Virtual Private Gateway*

vgw-08db0cb9576b8417b

Customer Gateway

☒ Existing

☐ New

Customer Gateway ID*

cgw-00cc6d655e6407a0b

Routing Options

☐ Dynamic (requires BGP)

☒ Static

Static IP Prefixes

IP Prefixes	Source	State
10.148.0.0/20	-	-

Add Another Rule

Tunnel Inside Ip Version

☒ IPv4

☐ IPv6

Local IPv4 Network Cidr

0.0.0.0/0

Remote IPv4 Network Cidr

0.0.0.0/0

- 下載在 AWS 中建立的通道設定檔。選取
- 供應商：一般
- 平台：一般
- 軟體：不限供應商
- IKE 版本：IKEv2
- 在 GCP 上建立 VPN 閘道和 VPN 通道。

← Google VPN gateway details DELETE

aws-vpn

Classic Cloud VPN gateway

VPC network	default
Region	asia-southeast1
IP address	34.126.140.162
High availability	No
Logs	View

請提供兩個通道的 IP 位址和 Ikev2 金鑰 (來自下載的 AWS 設定檔)。

- 完成後，AWS 和 GCP 上的通道應會處於「UP」和「Running」狀態。

Create VPN ConnectionDownload ConfigurationActions

search: 103Add filter1 to 1 of 1

Name	VPN ID	State	Virtual Private Gateway	Transit Gateway	Customer Gateway
ng-aws-gcp-mum-sg-tunnel-103	vpn-0aa4d244d1a87772f	available	vgw-08db0cb9576b8417b ng-...	-	cgw-00cc6d655e6407a0b

VPN Connection: vpn-0aa4d244d1a87772f

DetailsTunnel DetailsStatic RoutesTags

Tunnel State

Tunnel Number	Outside IP Address	Inside IPv4 CIDR	Inside IPv6 CIDR	Status	Status Last Changed
Tunnel 1	3.108.214.233	169.254.34.248/30	-	UP	January 28, 2022 at 12:35:13 PM
Tunnel 2	35.154.44.148	169.254.37.0/30	-	UP	January 28, 2022 at 12:38:29 PM

VPNVPN SETUP WIZARDREFRESHDELETESHOW INFO PANEL

CLOUD VPN TUNNELSCLOUD VPN GATEWAYSPEER VPN GATEWAYS

CREATE VPN TUNNEL

FilterEnter property name or value

	Tunnel name ↑	Cloud VPN gateway (IP)	Peer VPN gateway (IP)	Cloud Router BGP IP	BGP Peer IP	Routing type	VPN tunnel status	
☐	aws-vpn-tunnel (Classic)	aws-vpn 34.126.140.162	3.108.214.233	none	none	Route-based	Established	⋮
☐	aws-vpn-tunnel-2 (Classic)	aws-vpn 34.126.140.162	35.154.44.148	none	none	Route-based	Established	⋮

通道設定現已完成。

- 選取用於設定 Confluent Cloud 的 CIDR 區塊。在 AWS 上將此項目新增至 VPN 通道做為靜態路徑。

Name	VPN ID	State	Virtual Private Gateway	Transit Gateway	Customer Gateway
ng-aws-gcp-mum-sg-tunnel-103	vpn-0aa4d244d1a87772f	available	vgw-08db0cb9576b8417b ng-...	-	cgw-00cc6d655e6407a0b r

VPN Connection: vpn-0aa4d244d1a87772f

Details Tunnel Details **Static Routes** Tags

Edit

1 to 2 of 2

IP Prefixes	Source	State
192.168.0.0/16	-	available
10.148.0.0/20	-	available

注意：建立 AWS 客戶閘道時，請提供 GCP 保留 IP。

GCP Marketplace 上的 Confluent Kafka

- 從 GCP Marketplace 訂閱 Confluent Kafka。



Apache Kafka® on Confluent Cloud™

Confluent - Prod

Apache Kafka Re-Engineered for the Cloud

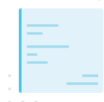
[MANAGE ON PROVIDER](#)

✓ [Purchased on 1/21/22](#)

- 登入 Confluent Cloud 並建立專屬叢集

Create cluster

1. Select cluster type ○ ○ ○ ○ ○



Basic

For basic or development use cases.
Upgrade to Standard at any time

Base cost: none

Uptime SLA: 99.5%

Usage limits

Ingress	up to 100 MB/s
Egress	up to 100 MB/s
Storage	up to 5,000 GB

Client connections	up to 1,000
Partitions	up to 2,048

Includes 10 free partitions



Standard

For production-ready use cases. Full
feature set and standard limits

Base cost: \$1.50 /hr

Uptime SLA: 99.95%

Usage limits

Ingress	up to 100 MB/s
Egress	up to 100 MB/s
Storage	up to 5,000 GB / Unlimited ⓘ

Client connections	up to 1,000
Partitions	up to 2,048

Includes 500 free partitions



Dedicated

For use cases with high traffic or that
require private networking

Price as sized 1 CKU / \$2.66 /hr

Uptime SLA: 99.95%



Ingress	up to 50 MB/s
Egress	up to 150 MB/s
Storage	up to 10,000 GB / Unlimited ⓘ

Client connections	up to 3,000
Partitions	up to 4,500

注意：Confluent Cloud 中的專屬叢集支援 VPC 對等互連。

Create cluster

1. Select cluster type — 2. Region/zones — 3. Networking 4. Security 5. Review and launch



Internet

Cluster will only be accessible over the public internet



VPC Peering

Cluster will only be accessible via VPC Peering

[See documentation](#)

Specify a CIDR block for your network region which will be used for allocating IP addresses and for IP routing. If you want to learn more, [read this doc](#).

CIDR for Confluent Cloud*

192.168.0.0/16

ⓘ This value may not be modified once your network region is provisioned

The CIDR block must be in one of the following private networks:

- 10.0.0.0/8
- 100.64.0.0/10
- 172.16.0.0/12
- 192.168.0.0/16

注意：Confluent 保證專屬叢集的運作時間達 99.95%。

- 提供 GCP 專案 ID，以及對等互連的虛擬私有雲網路詳細資料。

Cluster networking

Type

VPC peering

Cluster is available through VPC Peering connections

Network details

Confluent Cloud Cluster Network Details

- Confluent Cloud CIDR 192.168.0.0/16
- Confluent Cloud Project ID cc-prod-5
- Confluent Cloud VPC Network Name s-nrgdm

VPC Peering connections

	GCP Project ID	GCP Network Name	Status	Import custom routes
☐	raves-ce-vpcpeer	default	Active	Yes

[See documentation](#) on how to set up peering

- 使用提供的 Confluent Cloud 叢集網路詳細資料，在 GCP 端完成 VPC 對等互連。



✔ **confluent-peering**

Peer VPC network is connected

Your VPC network

default

Peered VPC network

s-nrgdm

Peered project ID

cc-prod-5

Exchange custom routes ?

Import & Export custom routes

Exchange subnet routes with public IP ?

Import & Export subnet routes with public IP

IMPORTED ROUTES

EXPORTED ROUTES

Routes in region

Region

asia-southeast1



- Confluent Cloud 叢集現已啟用，並與 GCP 對等互連。

Live (1)

 **raves_kafka**
✓ Running

Metrics

Production	Consumption	Storage
0B/s	0B/s	0B

Resources

Topics	ksqlDB	Connectors	Clients
0	0	0	0

Overview

ID	lkc-96pk7
Type	Dedicated
CKUs	1
Provider & region	GCP asia-southeast1

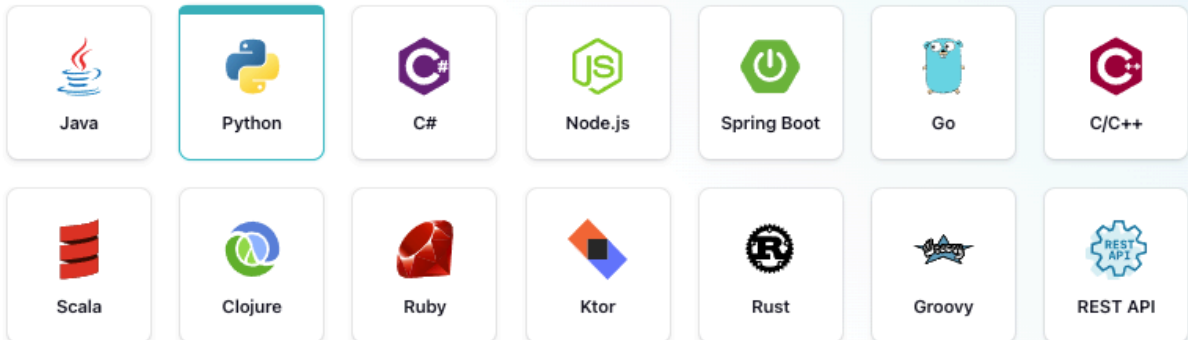
- 測試從 GCP 連線至 Confluent 叢集。
- 佈建 GCE 執行個體並安裝 Python3。
- 在 Confluent Cloud 中產生 Python 用戶端金鑰

New client

Produce and consume data with your cluster in the programming language of your choice.

1 Choose your language

Get the required configuration for your programming language.



2 Select configuration to copy into your client code (required)

Copy this configuration snippet into your client code to connect to this Confluent Cloud Kafka cluster.

Copy

```
1 # Kafka
2 bootstrap.servers=pkcs-p9gjm.asia-
3 southeast1.gcp.confluent.cloud:9092
4 security.protocol=SASL_SSL
5 sasl.mechanisms=PLAIN
6 sasl.username=[REDACTED]
7 sasl.password=[REDACTED]
8
9 # Best practice for higher availability in librdkafka clients
10 prior to 1.7
11 session.timeout.ms=45000
12
13 # Confluent Cloud Schema Registry
14 schema.registry.url=https://{ SR_ENDPOINT }
15 basic.auth.credentials.source=USER_INFO
16 basic.auth.user.info={{ SR_API_KEY }}:{{ SR_API_SECRET }}
```

Cluster API Key

An API key is required to connect to your cluster. You can either use an existing key or create a new one here.

Create Kafka cluster API key

- 從 GCP 執行個體執行測試腳本「producer.py」：

```

ravishgarg@python-inst:~/examples/clients/cloud/python$ ls -ltr
total 44
-rw-rw-r-- 1 ravishgarg ravishgarg 60 Jan 21 12:44 requirements.txt
-rwxrwxr-x 1 ravishgarg ravishgarg 3910 Jan 21 12:44 producer_ccsr.py
-rwxrwxr-x 1 ravishgarg ravishgarg 2542 Jan 21 12:44 producer.py
-rwxrwxr-x 1 ravishgarg ravishgarg 3976 Jan 21 12:44 consumer_ccsr.py
-rwxrwxr-x 1 ravishgarg ravishgarg 2720 Jan 21 12:44 consumer.py
-rwxrwxr-x 1 ravishgarg ravishgarg 5500 Jan 21 12:44 ccloud_lib.py
-rw-rw-r-- 1 ravishgarg ravishgarg 557 Jan 21 12:44 README.md
-rw-rw-r-- 1 ravishgarg ravishgarg 125 Jan 21 12:44 Dockerfile
drwxrwxr-x 2 ravishgarg ravishgarg 4096 Jan 28 07:33 __pycache__
-rw-rw-r-- 1 ravishgarg ravishgarg 517 Jan 28 07:39 python.config
ravishgarg@python-inst:~/examples/clients/cloud/python$ set -o vi
ravishgarg@python-inst:~/examples/clients/cloud/python$ python3 ./producer.py -f ./python.config -t test1
%4|1644238725.725|CONFWARN|rdkafka#producer-1| [thrd:app]: Configuration property session.timeout.ms is a consumer property and will
be ignored by this producer instance
%4|1644238725.733|CONFWARN|rdkafka#producer-2| [thrd:app]: Configuration property session.timeout.ms is a consumer property and will
be ignored by this producer instance
Producing record: alice {"count": 0}
Producing record: alice {"count": 1}
Producing record: alice {"count": 2}
Producing record: alice {"count": 3}
Producing record: alice {"count": 4}
Producing record: alice {"count": 5}
Producing record: alice {"count": 6}
Producing record: alice {"count": 7}
Producing record: alice {"count": 8}
Producing record: alice {"count": 9}
Produced record to topic test1 partition [0] @ offset 10
Produced record to topic test1 partition [0] @ offset 11
Produced record to topic test1 partition [0] @ offset 12
Produced record to topic test1 partition [0] @ offset 13
Produced record to topic test1 partition [0] @ offset 14
Produced record to topic test1 partition [0] @ offset 15
Produced record to topic test1 partition [0] @ offset 16
Produced record to topic test1 partition [0] @ offset 17
Produced record to topic test1 partition [0] @ offset 18
Produced record to topic test1 partition [0] @ offset 19
10 messages were produced to topic test1!

```

- 現在，請在 AWS 執行個體上，下載範例程式碼集，並執行「producer.py」指令碼，測試從 GCP 到 Confluent Cloud 的混合式連線能力。

```

ubuntu@ip-10-120-2-173:~/python$ python3 ./producer.py -f ./python.config -t test-aws
%4|1644239071.414|CONFWARN|rdkafka#producer-1| [thrd:app]: Configuration property session.timeout.ms is a consumer property and will
be ignored by this producer instance
%4|1644239071.422|CONFWARN|rdkafka#producer-2| [thrd:app]: Configuration property session.timeout.ms is a consumer property and will
be ignored by this producer instance
Topic test-aws created
Producing record: alice {"count": 0}
Producing record: alice {"count": 1}
Producing record: alice {"count": 2}
Producing record: alice {"count": 3}
Producing record: alice {"count": 4}
Producing record: alice {"count": 5}
Producing record: alice {"count": 6}
Producing record: alice {"count": 7}
Producing record: alice {"count": 8}
Producing record: alice {"count": 9}
Produced record to topic test-aws partition [0] @ offset 0
Produced record to topic test-aws partition [0] @ offset 1
Produced record to topic test-aws partition [0] @ offset 2
Produced record to topic test-aws partition [0] @ offset 3
Produced record to topic test-aws partition [0] @ offset 4
Produced record to topic test-aws partition [0] @ offset 5
Produced record to topic test-aws partition [0] @ offset 6
Produced record to topic test-aws partition [0] @ offset 7
Produced record to topic test-aws partition [0] @ offset 8
Produced record to topic test-aws partition [0] @ offset 9
10 messages were produced to topic test-aws!
ubuntu@ip-10-120-2-173:~/python$

```

i-076b744b697745102 (ng-vm-103-ubuntu)

Public IPs: 3.109.152.20 Private IPs: 10.120.2.173

程式碼範例：如要進行測試，可以從下列網址下載 Python 程式碼範例：

```
# git clone https://github.com/confluentinc/examples
```

```
# cd examples
```

```
# git checkout latest
```


步驟 3 · 約 0 分鐘

3. 恭喜

恭喜！您已成功在 AWS 和 GCP 之間建構多雲端跨區域的 Confluent Kafka 串流平台。

實用的程式碼研究室** ? **

查看一些程式碼研究室...

- [Google Compute Engine](#)

其他資訊

- [虛擬私有雲網路對接](#)
 - [Cloud VPN](#)
-